Seed Cleaner Grader



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 5 labs

Covers these lab items: Cleaner, grader; Seed Cleaner; Seed cleaning machines; Seed graders / seed separators

What it is used for

A seed cleaner grader removes unwanted materials from seed lots and separates seeds by size or other physical properties. Correct adjustment, clean handling, and regular inspection help improve seed quality and support consistent grading.

Main parts

- Feed hopper receives the uncleaned seed material.
- Feed gate controls the flow of seeds into the machine.
- Screens or sieves separate seeds by size.
- Air blower or aspiration system removes light impurities on equipped models.
- Vibrating or rotating mechanism moves seeds across the screens.
- Collection outlets direct graded seeds and rejected materials into separate containers.
- Drive motor and transmission power the cleaning mechanism.
- Control panel or adjustment levers regulate operating settings.

Before you start

1. Read the operating manual and confirm the machine is suitable for the seed type.
2. Inspect the hopper, screens, guards, belts, motor, and outlets for damage.
3. Ensure screens and air settings match the seed size and cleaning requirements.
4. Clean the machine to prevent mixing different seed varieties or batches.
5. Place labelled collection containers beneath the correct outlets.
6. Wear suitable PPE and keep the work area clear of loose materials.

How to use

1. Ensure the machine is switched off before changing screens or adjusting internal components.
2. Fit the recommended screens and secure all guards and covers.
3. Position labelled containers at the graded-seed and reject outlets.
4. Start the machine using the manufacturer's recommended sequence.
5. Feed a small quantity of seed and check the cleaning result.
6. Adjust feed rate, screen settings, or airflow within the permitted operating range.
7. Feed the remaining seed gradually without overloading the hopper.
8. Monitor seed flow, separation quality, unusual vibration, and dust during operation.
9. Stop the machine before clearing blockages or making adjustments that require access to moving parts.
10. Switch off and isolate the power, collect the output, and clean the machine after processing.

⚠ Safety

- Never put hands into the hopper, screens, or outlets while the machine is running.
- Keep guards fitted and do not bypass safety interlocks.
- Isolate power before removing blockages, changing screens, or cleaning internal parts.
- Use suitable respiratory protection and eye protection where seed dust is present.
- Keep loose clothing, jewellery, and hair away from belts and rotating parts.
- Do not operate the machine if the frame, motor, wiring, or guards are damaged.

Care & storage

- Remove seed residues and dust after each batch using approved cleaning methods.
- Inspect screens for tears, wear, and blockage before storage.
- Check belts, fasteners, bearings, and drive components according to the manual.
- Store the machine in a dry, clean area and record maintenance issues.

Common mistakes to avoid

- Using incorrect screens for the seed type or required grade.
- Feeding too quickly and overloading the machine.
- Allowing seed varieties or batches to mix between collection containers.
- Cleaning or clearing blockages while moving parts are active.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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